INTERNATIONAL DEVELOPMENT ASSOCIATION AND
INTERNATIONAL MONETARY FUND

MOZAMBIQUE

Joint World Bank/IMF Debt Sustainability Analysis

Prepared by the staffs of the International Development Association and
the International Monetary Fund

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This analysis assesses the sustainability of Mozambique's external public and domestic debt. The debt sustainability analysis (DSA) was conducted jointly by the staffs of the IMF and the World Bank using the joint Bank-Fund framework for debt sustainability analysis for low-income countries. The bilateral external debt data underlying this DSA were provided by the Mozambican authorities. The multilateral debt data provided by the authorities were reconciled with information obtained from the creditors. On the basis of this DSA, the staffs conclude that the risk of external debt distress for Mozambique is low.

I. BACKGROUND

1. **At end-2006, Mozambique's external public and publicly guaranteed (PPG) debt stock, including arrears,[1] stood at US$3.2 billion in nominal terms or US$706.6 million in NPV terms (Table 1).** Of this, 36 percent was owed to multilateral creditors, 59 percent to bilateral creditors, and about 5 percent to commercial creditors. Thanks to the Multilateral Debt Relief Initiative (MDRI), from the IMF, the International Development Association (IDA) of the World Bank, and the African Development Fund (AfDF), the debt stock fell by US$1.9 billion in nominal terms in 2006.[2]

[1] Arrears to bilateral creditors amounted to approximately US$1.2 billion, and include moratorium arrears to Portugal and Japan. Arrears to commercial creditors, US$157 million were covered by the debt-buy back facility.

[2] The amount of MDRI relief provided by the IMF was US$120.6 million; IDA provided US$1.3 billion; and AfDF provided US$464.5 million.

2. **Previously, Mozambique has benefited from HIPC assistance provided by multilateral and Paris Club bilateral creditors.**[3] Bilateral agreements were reached with all Paris Club creditors except Portugal and Japan, who together account for about 20 percent of total Paris Club debt at end-2000 in NPV terms.[4] The Mozambican authorities are making best efforts, however, to reach debt cancellation agreements with Portugal and Japan in 2007. Mozambique signed an agreement with France regarding the cancellation of all outstanding debts in April 2007. Russia recently announced its intention to cancel 100 percent of debt to HIPCs that reach completion point. Mozambique authorities are in touch with the Russian embassy in Maputo to finalize this process.

Table 1. Mozambique: External and Domestic Nominal Debt Outstanding at End-2006

	In millions of U.S. dollars	In percent of GDP
Total Domestic and External PPG Debt	**3833.3**	**49.6**
Total Domestic Debt	**606.9**	**7.8**
Total External Public and Privatly Guaranteed Deb	**3226.4**	**41.7**
Multilateral Creditors	**1149.0**	**14.9**
IDA	667.3	8.6
IMF	4.7	0.1
African Development Bank	187.1	2.4
Other Multilaterals	289.9	3.7
Bilateral Creditors 1/	**1902.0**	**24.6**
Paris Club	801.2	10.4
Non-Paris Club	1100.9	14.2
Commercial Creditors 1/	**175.4**	**2.3**
Total Private Non-guaranteed and Short Term Debt	**1404.6**	**18.2**
Memo items:		
Total External Debt	**4631.0**	**59.9**
NPV of Total External Debt	**2111.2**	**27.3**
NPV of External PPG Debt	**706.6**	**9.1**

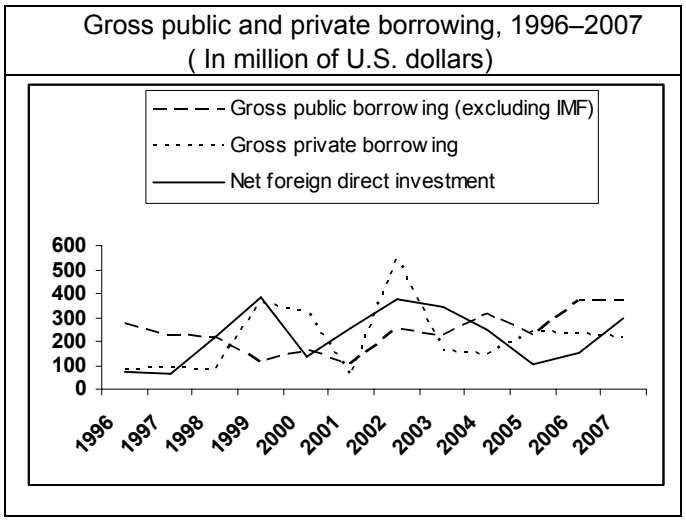
[3] See "Mozambique−HIPC Debt Initiative: President's Memorandum and Recommendation and Completion Point Document" (IDA/R99–139), and "Mozambique – Enhanced HIPC Debt Initiative: President's Memorandum and Recommendation and Completion Point Document" (IDA/R2001–0150).

[4] In February 2005, Mozambique signed a technical memorandum with Portugal, in which Portugal confirmed that it would provide 100 percent debt service reduction. Subsequently, however, Portugal added a clause to this agreement which Mozambique has refused to accept, because it considers it to be not on comparable terms. Japan, has also repeatedly indicated intentions for 100 percent debt reduction, but an agreement has not yet been signed due to delays of an administrative nature.

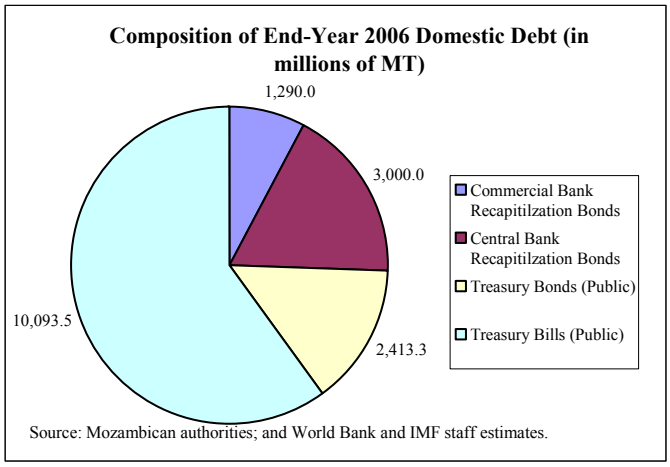
3. **Thus far, Mozambique has not been that successful in negotiating debt relief agreements with some of its non-Paris Club bilateral creditors.** Negotiations with Algeria, Libya, and Iraq are proving difficult despite Mozambique's best efforts. Negotiations with India are reportedly progressing slowly. The Chinese government cancelled Mozambique's outstanding debt in February 2007 and Kuwait provided comparable treatment with the Paris Club.

4. **Mozambique benefited from a commercial debt buy-back operation in 2007.** This operation covered all outstanding commercial debt in arrears, which was estimated at US$175.4 million at end-2006. The Government of Norway and the World Bank (IBRD) financed the operation.

5. **Total external debt in Mozambique has a significant private component due to "megaproject" related lending.** Private external debt averaged 24 percent of the country's total external debt during the period 2001–06. Most of the megaproject investments were debt-creating, as evidenced by the correlation between FDI inflows and private borrowing (text chart).[5]



6. **Mozambique's domestic public debt stock is dominated by recapitalization bonds and treasury bills issued to mop up excess liquidity.** The total stock of domestic public debt (treasury bills and bonds) was about 8 percent of GDP at end-2006, which is low by regional levels. The Ministry of Finance ceased issuing treasury bills for deficit financing purposes in 2006, therefore the total stock of outstanding treasury bills have been issued by the Bank of



[5] Gross private borrowing includes gross private external borrowing by megaprojects and other private companies, including net borrowing by commercial banks from 2002, but is based on the limited information available and may not cover all private enterprises.

Mozambique on behalf of government for sterilization purposes. The majority of the stock of treasury bonds are accounted for by bonds issued to recapitalize the central bank in 2005 and 2006 (following losses incurred during 2004), and related to the process of restructuring commercial banks, which is now complete. The rest of the treasury bonds with a maturity ranging from 1–10 years are held by the general public and financial sector.

II. METHODOLOGY AND KEY ASSUMPTIONS

7. **Following the guidelines of the LIC debt sustainability framework, staffs have analyzed the evolution of the external public debt stock and debt service indicators for Mozambique under a baseline scenario and a series of stress tests.**[6] The stress tests are designed to assess the probability of a country facing debt distress in the future under common shocks. In addition, staffs have considered the effect of an increase in nonconcessional external new borrowing by US$1 billion in 2008[7] and a persistent scaling-up of external borrowing of US$200 million from 2008–15.[8] The purpose of the first country-specific scenario is to illustrate the potential impact of undertaking large infrastructure projects that increase the central government's liabilities to commercial creditors. The scaling-up scenario is intended to illustrate the fiscal space available to scale-up concessional external borrowing focused on achieving the Millennium Development Goals (MDGs).[9]

8. **The analysis is guided by indicative, performance-based debt burden thresholds, which take into account the empirical finding that the debt levels that a low-income country can sustain increase with the quality of its policies and institutions.** The quality of policies and institutions is measured by the three-year average Country Policy and Institutional Assessment (CPIA) of the World Bank, according to which Mozambique ranks as a 'medium performer'. The indicative external debt burden thresholds for countries in this category are an NPV of debt-to-exports ratio of 150 percent, NPV of debt-to-revenue ratio of 250 percent, a NPV of debt-to-GDP ratio of 40 percent, and debt-service-to-exports and revenues ratios of 20 and 30 percent, respectively.[10]

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[6] The Executive Boards of the Fund and the Bank approved the Operational Framework for Debt Sustainability Assessments in Low-Income Countries in April 2005 (http://www.imf.org/External/np/pp/eng/2005/032805.htm) and IDA/R2005-0056).

[7] The assumed terms of this loan are a grace period of one year, total maturity of 10 years, and an interest rate of 8 percent.

[8] External borrowing from bilateral donors is assumed to be higher by US$200 million each year from 2008 onwards until 2015.

[9] It is also based on the indication from donors of a possible further scaling-up of foreign aid of about US$200 million (or 2 percent of 2008 GDP) over the medium term, which is unlikely to be all grants.

[10] The NPV of external debt-to-revenue ratio excludes all grants.

9. **The baseline scenario is subject to a number of assumptions.** The underlying macroeconomic assumptions, which are summarized in Box 1, are consistent with the medium-term macroeconomic framework described in the staff report. In addition, the external debt numbers assume the full delivery of HIPC debt relief by all creditors, 100 percent debt reduction by France, Russia, Japan China, and completion of the commercial debt buyback operation. Finally, the authorities are assumed to borrow predominantly from the IDA and AfDF, resulting in an average grant element on new borrowing of 41 percent. This assumption is based on the authorities' medium-term projections, which were informed by consultations with donors.[11]

III. External Debt Sustainability

10. **Under the baseline scenario presented in Table 2, the debt indicators all remain below their respective thresholds.** The NPV of PPG external debt-to-GDP ratio is projected to rise from 11.6 percent in 2007 to 16.1 percent by 2013, after which it slowly declines over the remainder of the projection period to 9.9 percent by 2027. It thus remains well below the country-specific threshold of 40 percent. The NPV of PPG debt-to-exports ratio increases from 30.3 percent in 2007 to 63.3 percent by 2016—also far below the threshold of 150 percent—before falling back to about 44 percent again by 2027. The NPV of PPG debt-to-revenue ratio increases to a peak of 94.5 percent in 2010, significantly below the threshold of 250 percent. It then declines rapidly to 35.9 percent by 2027, driven in part by the assumed increase in revenue collection.

11. **The debt service indicators also remain below their thresholds under the baseline.** As a result of the assumed full delivery of remaining HIPC debt relief in 2007, PPG debt service falls in 2007. The PPG debt service-to-exports ratio, which has a threshold at 20 percent, falls from 2.5 percent in 2006 to 1.1 percent in 2007, peaking at 4.6 percent in 2021 and then gradually decreasing to 3.8 percent in 2027, always well below the threshold. The ratio of PPG debt service to fiscal revenues falls from 6.6 percent in 2006, to 2.9 percent in 2007, increasing slowly to 5.1 percent in 2011 and declining to 3.1 by 2027, well below the 30-percent threshold (Table 2).

12. **External debt sustainability appears resilient to a number of shocks, although the DSA shows that the debt service burden could rise significantly in the event of a large increase in nonconcessional external commercial borrowing.**[12] As Table 3 shows,

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[11] Staffs have undertaken additional analysis that shows that the results of the DSA are not sensitive to a lowering of the grant element. Stress Test A2 underscores this finding.

[12] Both, last year's DSA and this one, concluded that the risk of external debt distress for Mozambique is low, even with the inclusion of the domestic debt, which was absent in the 2006 DSA. However, some of the expected bilateral agreements envisaged in the 2006 DSA did not materialized but are expected to be concluded during the course of this year. Regarding the assumptions for MDRI, the expected relief was greater than previously estimated, which further improved the debt situation.

all the stress tests results show that debt sustainability indicators are well below the relevant thresholds. However, stress test B7 shows that debt sustainability may be susceptible to a significant increase in the amount of nonconcessional commercial debt in 2008. While still remaining below the thresholds, the debt burden indicators rise sharply, and more than double compared to the baseline (Table 3 and Figure 1). The alternative scenario of higher concessional borrowing from 2008–15 also remains below the indicative thresholds but almost doubles the NPV of debt to GDP by 2015.

IV. PUBLIC SECTOR DEBT SUSTAINABILITY

13. **Mozambique's public debt (including domestic debt) is expected to decline over the medium-to-long term under the baseline scenario (Table 2a and Figure 1a).** As a result of the gradual redemption of bonds issued to restructure the banking system and predominant use of foreign exchange sales for sterilization purposes, public debt is projected to decline from about 27 percent of GDP in 2007 to 30 percent in 2011, and to 14 percent by 2027 while the NPV of public sector debt-to GDP ratio is projected to fall from 19 percent in 2007 to 9 percent by 2027 (Table 2a). The nonrecourse to domestic financing over the medium-term whilst only gradually increasing domestic financing of the fiscal deficit from 2012 onwards contributes to such a path for public debt. On the other hand, this projection rests on the assumption that the revenue to GDP ratio increases and foreign aid gradually falls as a percentage of GDP (Box 1), requiring only a modest increase in domestic debt financing to maintain a roughly unchanged level of government expenditure.

14. **Public sector debt dynamics are driven by the external debt, given the low and declining level of domestic debt** (Table 1a). The large proportion of external debt makes the public debt projection vulnerable to the same set of shocks as the external debt sustainability assessment, albeit with some additional risks related to the sterilization needs to mop up excess liquidity; which would increase in the event of lower revenues and/or higher expenditures (even if financed by higher aid), and aid volatility and/or reluctance to sell foreign exchange due to possible Dutch-Disease effects. The stress tests indicate that public sector debt ratios are most vulnerable to a permanently lower GDP growth, which shows an increasing path of public debt.

Box 1. Macroeconomic Assumptions 2007–27

The medium-term assumptions in the baseline scenario for 2007–27 are consistent with the authorities' medium-term macroeconomic framework described in the government's Plano de Acção para Redução da Pobreza Absoluta II (PARPA II).

Real GDP growth is projected to average 7.0 percent per year during 2007–10 and 6.5 percent thereafter, representing a deceleration from its annual average of 8.6 percent during 2001–06. The 6.5–7.0 percent projection falls within the 5–8 percent range projected by the World Bank in its most recent Country Economic Memorandum for Mozambique (Report No. 32615-MZ).

Consumer price inflation is projected to fall to 5 percent in the medium-to-long term, as oil prices stabilize and inflation converges to levels closer to Mozambique's main trading partner (South Africa).

The real effective exchange rate is expected to remain broadly unchanged, in line with the relative stability observed over the past decade.

Export growth is projected to accelerate slowly from 5.5 percent in the medium term to 6.1 percent in the long term. This is driven to a large extent by traditional (nonmegaproject) exports that grow at the rate of Mozambique's trade partner import demand growth and megaproject exports that are affected by world prices on aluminum and gas.

Import growth is projected to decelerate slowly from 6.6 percent in medium term to 6.3 percent in the long-term. Import growth associated with megaprojects is assumed to equal the rate of megaproject export growth, assuming that the import content of megaprojects remains roughly constant on average. All other imports are assumed to grow at the rate of real GDP growth.

The noninterest current account deficit after grants is projected to expand slowly from 3.2 percent of GDP during 2007–12 to 3.7 percent in 2013–2026, due to slightly higher import growth than export growth.

Fiscal revenues are expected to rise from 14 percent in 2006 to 16 percent of GDP in 2010 reflecting a 0.5 percent of GDP annual revenue effort on account of improved revenue administration and broadening of the tax base. Beyond 2010, nontax revenues from natural resource exploitation particularly megaprojects are expected to make a growing fiscal contribution. The tax to GDP ratio is projected to reach about 22 percent in the long-term; the level estimated as Mozambique's potential tax ratio by a number of studies.[13] Total expenditures remain roughly constant as a percent of GDP in the long term, resulting in a domestic primary deficit ranging between 2–4 percent of GDP from 2007–27.

External grants to the government are projected to remain high at an average of 9 percent of GDP during the medium term, which will help finance the "second wave of reforms" needed to sustain broad-based growth and achieve the MDGs by 2015. No grant financing is assumed to come forward from IDA. The share of program and project grants versus lending is assumed to fall over time: grants are assumed to fall from approximately 8 percent of GDP in 2006 to around 2.3 percent of GDP by 2027.

[13] IMF Country Report No. 05/311.

V. CONCLUSIONS

15. **In staffs' view, Mozambique faces a low risk of debt distress.** Mozambique's external debt levels are expected to remain well below their indicative thresholds for debt distress both under the baseline and under various stress tests. Mozambique's public debt is expected to decline over the medium-to-long term under the baseline, while the stress tests indicate that public sector debt ratios are most vulnerable to a permanently lower GDP growth.

16. **A prudent new borrowing policy remains important.** Where possible, the authorities should continue to rely on concessional borrowing and grants to minimize future debt service, and any nonconcessional financing of new projects ought to be considered case by case based on economic return, impact on debt sustainability, and potential effects on the financing decisions of donors and concessional lenders. Also, the authorities would still need to closely monitor any substantial scaling-up of concessional external borrowing and the issuance of domestic debt to sterilize the additional liquidity injected from increased government spending.

Table 2. Mozambique: External Debt Sustainability Framework, Baseline Scenario, 2004–27 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 7/	Standard Deviation 7/	Projections								
	2004	2005	2006			2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027
External debt (nominal) 1/	**96.0**	**89.5**	**59.8**			**36.5**	**36.8**	**37.4**	**37.6**	**37.1**	**36.6**		**33.0**	**23.6**
o/w public and publicly guaranteed (PPG)	73.9	69.5	41.7			20.1	22.5	23.9	24.7	25.3	25.4		23.6	14.9
Change in external debt	-16.2	-6.6	-29.6			-23.4	0.4	0.5	0.2	-0.4	-0.5		-0.8	-0.8
Identified net debt-creating flows	-20.2	-5.4	-32.1			-1.4	0.3	0.9	1.6	-4.2	-3.0		-0.6	1.8
Non-interest current account deficit	**2.7**	**7.0**	**-21.6**	**3.8**	**9.9**	**3.8**	**3.6**	**4.6**	**5.8**	**0.2**	**1.3**	**3.2**	**3.3**	**5.1**
Deficit in balance of goods and services	9.4	10.7	6.1			7.9	9.3	10.6	11.2	6.9	7.5		8.1	6.3
Exports	31.0	31.7	37.3			38.1	34.9	32.1	30.5	33.0	30.6		24.4	22.4
Imports	40.4	42.4	43.5			46.1	44.3	42.8	41.7	39.9	38.2		32.5	28.7
Net current transfers (negative = inflow)	-8.9	-6.8	-33.2	-12.6	7.7	-11.7	-10.9	-10.1	-9.2	-10.4	-9.7		-7.7	-3.6
o/w official	-8.9	-6.8	-33.2			-11.7	-10.9	-10.1	-9.2	-10.4	-9.7		-7.7	-3.6
Other current account flows (negative = net inflow)	2.2	3.1	5.5			7.6	5.2	4.1	3.8	3.6	3.4		2.9	2.5
Net FDI (negative = inflow)	**-4.1**	**-1.5**	**-2.0**	**-5.1**	**3.0**	**-3.7**	**-3.4**	**-3.7**	**-3.8**	**-3.7**	**-3.6**		**-3.1**	**-2.8**
Endogenous debt dynamics 2/	**-18.7**	**-10.9**	**-8.6**			**-1.6**	**0.1**	**-0.1**	**-0.4**	**-0.7**	**-0.7**		**-0.8**	**-0.6**
Contribution from real GDP growth	-6.8	-5.2	-6.7			-4.0	-2.4	-2.3	-2.4	-2.2	-2.2		-2.0	-1.5
Contribution from price and exchange rate changes	-14.4	-7.7	-3.9			…	…	…	…	…	…		…	…
Residual (3-4) 3/	**4.0**	**-1.2**	**2.5**			**-21.9**	**0.1**	**-0.4**	**-1.4**	**3.7**	**2.4**		**-0.1**	**-2.5**
NPV of external debt 4/	…	…	27.3			27.9	27.6	27.8	27.9	27.5	27.1		24.8	18.5
In percent of exports	…	…	73.1			73.3	79.0	86.6	91.5	83.6	88.5		101.6	82.7
NPV of PPG external debt 5/	**…**	**…**	**9.1**			**11.6**	**13.2**	**14.3**	**15.1**	**15.7**	**16.0**		**15.3**	**9.9**
In percent of exports	**…**	**…**	**24.5**			**30.3**	**37.9**	**44.5**	**49.5**	**47.6**	**52.1**		**63.0**	**44.0**
In percent of government revenues	**…**	**…**	**65.4**			**79.7**	**88.5**	**92.3**	**94.5**	**93.9**	**91.7**		**73.0**	**36.0**
Debt service-to-exports ratio (in percent)	**23.9**	**18.0**	**13.9**			**16.2**	**17.7**	**17.8**	**17.2**	**18.3**	**18.7**		**18.1**	**15.6**
PPG debt service-to-exports ratio (in percent)	**3.8**	**4.3**	**2.5**			**1.1**	**1.2**	**1.4**	**1.6**	**1.5**	**2.3**		**3.8**	**3.8**
PPG debt service-to-revenue ratio (in percent)	**9.3**	**10.0**	**6.6**			**2.9**	**2.8**	**2.9**	**3.0**	**3.0**	**4.0**		**4.4**	**3.1**
Total gross financing need (billions of U.S. dollars)	0.4	0.8	-1.4			0.5	0.6	0.6	0.8	0.3	0.4		0.9	2.5
Non-interest current account deficit that stabilizes debt ratio	18.9	13.6	8.0			27.2	3.2	4.1	5.6	0.6	1.8		4.0	5.9
Key macroeconomic assumptions														
Real GDP growth (in percent)	7.5	6.2	8.5	8.2	3.3	7.0	7.0	7.0	7.0	6.5	6.5	6.8	6.5	6.5
GDP deflator in US dollar terms (change in percent)	14.7	8.8	4.5	1.5	9.0	-1.8	1.4	3.0	3.0	2.9	2.9	1.9	2.0	1.0
Effective interest rate (percent) 6/	2.7	2.5	2.5	2.9	0.6	4.2	7.4	6.8	5.8	4.6	4.4	5.5	4.1	3.8
Growth of exports of G&S (US dollar terms, in percent)	35.1	18.4	33.6	20.2	11.7	7.3	-0.6	1.4	4.7	18.4	2.0	5.5	7.1	6.9
Growth of imports of G&S (US dollar terms, in percent)	10.4	21.4	16.2	13.1	11.0	11.4	4.2	6.5	7.4	4.9	5.0	6.6	6.6	6.5
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	41.2	40.9	40.6	40.5	40.4	40.4	40.7	40.4	40.4
Government revenues (excluding grants, in percent of GDP)	12.6	13.6	14.0	12.6		14.5	15.0	15.5	16.0	16.7	17.4	15.8	21.0	27.4
Grant-equivalent financing (in percent of GDP) 8/	…	…	…	…		13.5	12.4	11.3	10.4	9.6	8.8	11.0	5.6	2.7
Grant-equivalent financing (in percent of external financing) 8/	…	…	…	…		82.8	83.5	84.0	84.3	84.6	84.3	83.9	83.0	83.1
Memorandum item:														
Nominal GDP (billions of US dollars)	5.9	6.8	7.7			8.1	8.8	9.7	10.7	11.7	12.9		20.0	42.6
Nominal dollar GDP growth	23.3	15.6	13.4			5.1	8.5	10.2	10.2	9.6	9.6		8.6	7.5
NPV of PPG external debt (in billions of US dollars)			0.7			0.9	1.2	1.4	1.6	1.8	2.1		3.1	4.2
(NPVt-NPVt-1)/GDPt-1 (in percent)						3.0	2.8	2.5	2.3	2.1	1.8	2.4	1.0	0.2

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); the full delivery of the HIPC debt releif by all creditors; 100 percent debt reduction by France, Russia, Japan, and China; completion of commercial debt buy-back operation; changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ NPV for 2006 excludes arrears.
6/ Current-year interest payments divided by previous period debt stock.
7/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
8/ Grant financing includes grants provided directly to the government and through new borrowing (difference between the face value and the NPV of new debt).

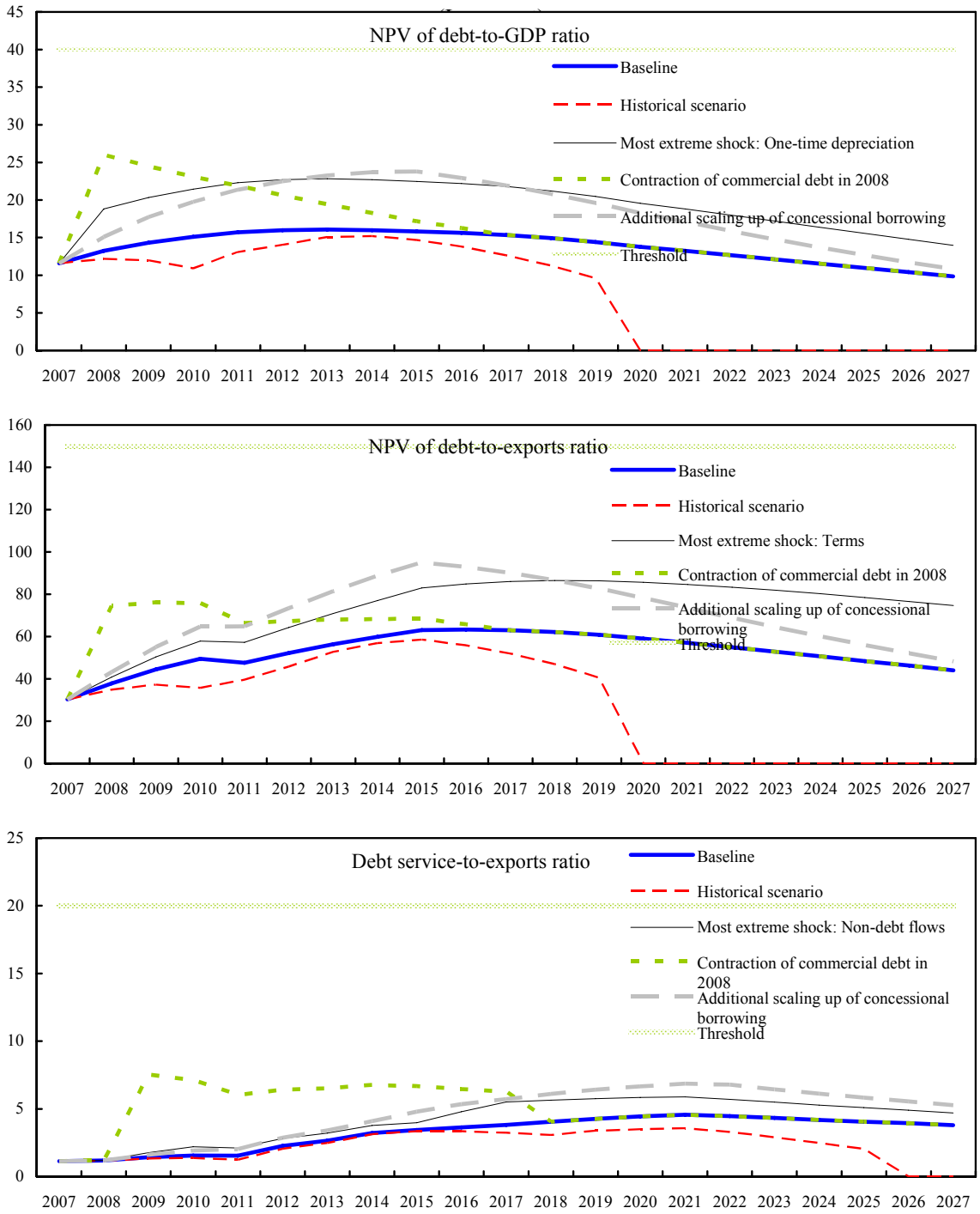
Table 3. Mozambique: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2007-27
(In percent)

		Projections						
	2007	2008	2009	2010	2011	2012	**2017**	2027
NPV of debt-to-GDP ratio								
Baseline	12	13	14	15	16	16	**15**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	12	12	12	11	13	14	**13**	-7
A2. New public sector loans on less favorable terms in 2008-27 2/	12	14	16	18	19	20	**21**	17
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	12	14	15	16	16	17	**16**	10
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	12	11	9	10	11	12	**12**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	12	15	17	18	19	19	**19**	12
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	12	18	23	23	23	23	**20**	12
B5. Combination of B1-B4 using one-half standard deviation shocks	12	12	9	11	12	12	**13**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	12	19	20	21	22	23	**22**	14
B7. Contraction of commercial debt	12	26	25	23	22	21	15	10
B8. Additional scaling up of concessional borrowing	12	15	18	20	21	22	22	11
NPV of debt-to-exports ratio								
Baseline	30	38	45	49	48	52	**63**	44
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	30	35	37	36	40	46	**52**	-31
A2. New public sector loans on less favorable terms in 2007-26 2/	30	41	50	58	57	64	**86**	75
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	30	38	45	49	48	52	**63**	44
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	30	30	24	29	29	32	**43**	33
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	30	38	45	49	48	52	**63**	44
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	30	51	70	75	70	75	**84**	52
B5. Combination of B1-B4 using one-half standard deviation shocks	30	29	20	24	24	27	**37**	29
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	30	38	45	49	48	52	**63**	44
B7. Contraction of commercial debt	30	74	76	76	66	67	63	44
B8. Additional scaling up of concessional borrowing	30	43	55	65	65	73	90	48
NPV of debt-to-revenue ratio								
Baseline	80	88	92	95	94	92	**73**	36
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	80	82	77	68	78	81	**60**	-26
A2. New public sector loans on less favorable terms in 2007-26 2/	80	95	104	111	113	113	**100**	61
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	80	90	96	98	98	95	**76**	37
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	80	75	59	64	66	67	**58**	32
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	80	97	113	115	115	112	**89**	44
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	80	118	146	144	138	132	**97**	42
B5. Combination of B1-B4 using one-half standard deviation shocks	80	82	60	66	69	70	**62**	35
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	80	126	131	134	133	130	**104**	51
B7. Contraction of commercial debt	80	174	158	145	131	118	73	36
B8. Additional scaling up of concessional borrowing	80	101	114	124	128	129	104	39
Debt service-to-exports ratio								
Baseline	1	1	1	2	2	2	**4**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	1	1	1	1	1	2	**3**	1
A2. New public sector loans on less favorable terms in 2008-27 2/	1	1	1	2	2	2	**4**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	1	1	1	2	2	2	**4**	4
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	1	1	1	1	1	2	**2**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	1	1	1	2	2	2	**4**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	1	1	2	2	2	3	**6**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	1	1	1	1	1	1	**2**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	1	1	1	2	2	2	**4**	4
B7. Contraction of commercial debt	1	1	8	7	6	6	6	4
B8. Additional scaling up of concessional borrowing	1	1	2	2	2	3	6	5
Debt service-to-revenue ratio								
Baseline	3	3	3	3	3	4	**4**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008-27 1/	3	3	3	3	2	4	**4**	1
A2. New public sector loans on less favorable terms in 2008-27 2/	3	3	3	3	4	4	**5**	4
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008-09	3	3	3	3	3	4	**5**	3
B2. Export value growth at historical average minus one standard deviation in 2008-09 3/	3	3	3	2	2	3	**3**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008-09	3	3	4	4	4	5	**5**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2008-09 4/	3	3	4	4	4	5	**6**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	3	3	3	2	3	4	**3**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2008 5/	3	4	4	4	4	6	**6**	4
B7. Contraction of commercial debt	3	3	16	14	12	11	7	3
B8. Additional scaling up of concessional borrowing	3	3	3	4	4	5	7	4
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	39	39	39	39	39	39	**39**	39

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 1. Mozambique: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2007-2027



Source: Staff projections and simulations.

Table 1a.Mozambique: Public Sector Debt Sustainability Framework, Baseline Scenario, 2004-2027
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2004	2005	2006	Historical Average 5/	Standard Deviation 5/	2007	2008	2009	2010	2011	2012	2007-12 Average	2017	2027	2013-27 Average
Public sector debt 1/	68.5	81.7	52.0			26.9	27.8	28.7	29.1	29.7	29.7		25.5	14.5	
o/w foreign-currency denominated	60.5	73.7	43.9			19.9	22.2	24.8	25.8	25.9	26.3		24.1	13.2	
Change in public sector debt	-18.9	13.2	-29.7			-25.1	0.9	0.9	0.4	0.6	0.0		-0.5	-1.2	
Identified debt-creating flows	-25.6	5.6	-32.5			1.3	0.6	0.0	-0.5	-0.5	-0.7		0.2	0.5	
Primary deficit	3.3	1.3	0.5	2.8	2.1	3.4	2.6	2.1	1.6	1.4	1.3	2.1	2.0	1.7	2.1
Revenue and grants	20.2	20.0	23.9			26.0	25.7	25.4	25.1	25.1	25.1		25.4	24.7	
of which : grants	7.5	6.3	9.9			11.5	10.7	9.9	9.1	8.4	7.7		4.8	1.9	
Primary (noninterest) expenditure	23.5	21.3	24.4			29.4	28.2	27.4	26.7	26.5	26.4		27.4	26.4	
Automatic debt dynamics	-26.7	4.8	-10.0			-2.1	-1.9	-2.0	-2.1	-1.8	-1.9		-1.8	-1.1	
Contribution from interest rate/growth differential	-7.0	-5.3	-7.3			-3.4	-1.6	-1.7	-1.9	-1.6	-1.7		-1.5	-1.0	
of which : contribution from average real interest rate	-0.9	-1.2	-0.9			0.0	0.2	0.1	0.0	0.1	0.1		0.0	-0.1	
of which : contribution from real GDP growth	-6.1	-4.0	-6.4			-3.4	-1.8	-1.8	-1.9	-1.8	-1.8		-1.6	-1.0	
Contribution from real exchange rate depreciation	-19.7	10.0	-2.8			1.3	-0.4	-0.3	-0.2	-0.2	-0.2		...	...	
Other identified debt-creating flows	-2.2	-0.5	-23.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-1.9	-0.3	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-0.4	-0.2	-23.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	6.7	7.6	2.8			-26.4	0.3	0.9	0.9	1.1	0.7		-0.7	-1.7	
NPV of public sector debt	8.0	18.3	17.7			19.0	19.1	18.5	18.7	19.8	19.7		16.7	9.6	
o/w foreign-currency denominated	0.0	10.3	9.5			12.0	13.5	14.6	15.4	16.0	16.3		15.3	8.4	
o/w external	...	10.3	9.5			12.0	13.5	14.6	15.4	16.0	16.3		15.3	8.4	
NPV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	5.0	2.7	2.1			4.6	3.4	2.8	2.5	2.4	2.2		2.8	2.3	
NPV of public sector debt-to-revenue and grants ratio (in percent)	39.5	91.7	74.2			72.9	74.6	72.9	74.6	78.7	78.5		65.7	39.0	
NPV of public sector debt-to-revenue ratio (in percent)	...	134.4	126.7			130.6	127.9	119.3	117.1	118.4	113.0		80.8	42.3	
o/w external 3/	...	75.7	68.3			82.8	90.3	94.1	96.2	95.6	93.3		74.3	36.7	
Debt service-to-revenue and grants ratio (in percent) 4/	7.4	6.3	5.3			4.7	3.3	3.0	3.3	4.2	3.9		3.3	2.4	
Debt service-to-revenue ratio (in percent) 4/	11.8	9.3	9.1			8.4	5.6	4.9	5.2	6.3	5.6		4.0	2.6	
Primary deficit that stabilizes the debt-to-GDP ratio	22.2	-11.9	30.2			28.6	1.6	1.2	1.3	0.8	1.3		2.5	2.9	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.5	6.2	8.5	8.5	3.3	7.0	7.0	7.0	7.0	6.5	6.5	6.8	6.5	6.5	6.5
Average nominal interest rate on forex debt (in percent)	0.4	0.6	0.4	0.4	0.2	0.5	1.4	1.4	1.5	1.6	1.6	1.3	1.5	1.3	1.4
Average real interest rate on domestic currency debt (in percent)	11.1	-3.3	-5.3	-4.9	12.3	4.8	2.5	2.5	3.3	8.2	5.1	4.4	15.8	1.6	4.9
Real exchange rate depreciation (in percent, + indicates depreciation)	-25.7	17.9	-4.1	2.2	15.6	3.2	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	8.9	11.1	13.2	10.3	4.4	6.4	5.7	5.4	5.1	5.0	5.0	5.4	5.0	5.0	5.0
Growth of real primary spending (deflated by GDP deflator, in percent)	-0.5	-3.7	24.4	8.9	14.3	29.1	2.6	4.0	4.1	5.9	6.0	8.6	6.4	5.9	6.5

Sources: Country authorities; and Fund staff estimates and projections.

1/ General government gross debt.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

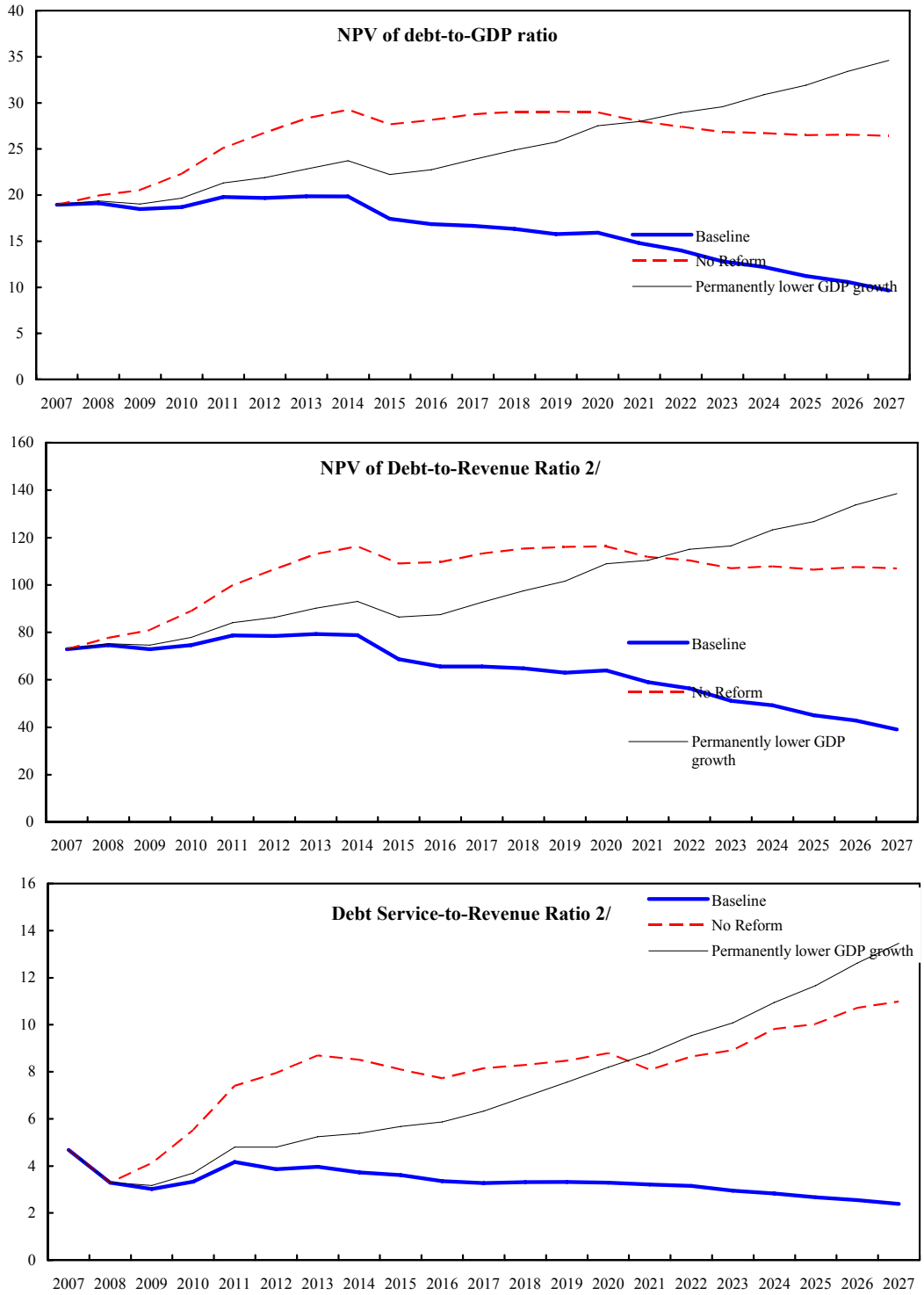
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2a.Mozambique: Sensitivity Analysis for Key Indicators of Public Debt 2007-2027

	Projections							
	2007	2008	2009	2010	2011	2012	2017	2027
NPV of Debt-to-GDP Ratio								
Baseline	19	19	18	19	20	20	17	10
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	19	19	19	20	22	23	22	19
A2. Primary balance is unchanged from 2007	19	20	21	22	25	27	29	26
A3. Permanently lower GDP growth 1/	19	19	19	20	21	22	24	35
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	19	20	20	21	22	23	22	18
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	19	21	23	23	24	24	20	12
B3. Combination of B1-B2 using one half standard deviation shocks	19	20	21	21	22	22	19	11
B4. One-time 30 percent real depreciation in 2008	19	23	22	21	22	22	18	11
B5. 10 percent of GDP increase in other debt-creating flows in 2008	19	28	27	27	28	27	23	13
NPV of Debt-to-Revenue Ratio 2/								
Baseline	73	75	73	75	79	78	66	39
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	73	75	76	81	90	94	90	80
A2. Primary balance is unchanged from 2007	73	78	81	89	100	107	113	107
A3. Permanently lower GDP growth 1/	73	75	75	78	84	86	93	139
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	73	76	77	81	87	89	85	74
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	73	83	92	93	96	95	79	47
B3. Combination of B1-B2 using one half standard deviation shocks	73	79	84	85	89	88	73	43
B4. One-time 30 percent real depreciation in 2008	73	92	85	85	88	86	71	45
B5. 10 percent of GDP increase in other debt-creating flows in 2008	73	111	107	107	110	108	89	53
Debt Service-to-Revenue Ratio 2/								
Baseline	5	3	3	3	4	4	3	2
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	5	3	3	4	6	7	6	8
A2. Primary balance is unchanged from 2007	5	3	4	6	7	8	8	11
A3. Permanently lower GDP growth 1/	5	3	3	4	5	5	6	13
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2008-2009	5	3	3	4	5	5	5	6
B2. Primary balance is at historical average minus one standard deviations in 2008-2009	5	3	6	8	6	5	4	4
B3. Combination of B1-B2 using one half standard deviation shocks	5	3	5	6	5	5	4	3
B4. One-time 30 percent real depreciation in 2008	5	3	4	4	5	5	4	3
B5. 10 percent of GDP increase in other debt-creating flows in 2008	5	3	16	8	7	6	5	5

Sources: Country authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

Figure 1a. Mozambique: Indicators of Public Debt Under Alternative Scenarios, 2007-2027 1/



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2017.
2/ Revenue including grants.